

Candidate Pseudolabel Learning: Enhancing Vision-Language Models by Prompt Tuning with Unlabeled Data

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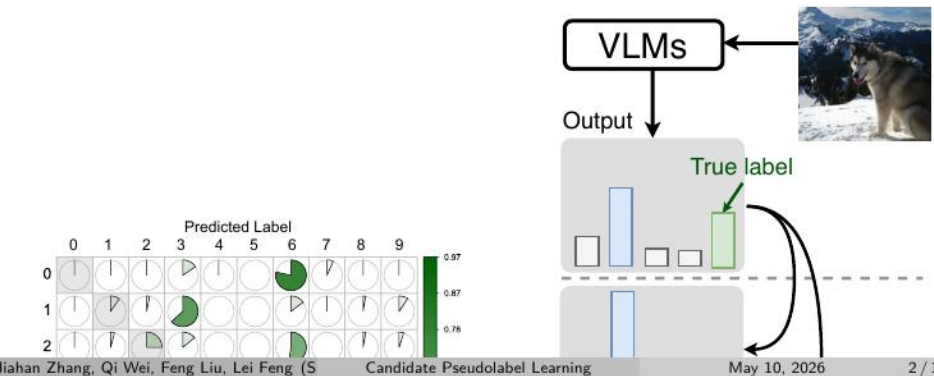
Instructions

- Slides summarize the core ideas, math, and results from the paper.

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Motivation

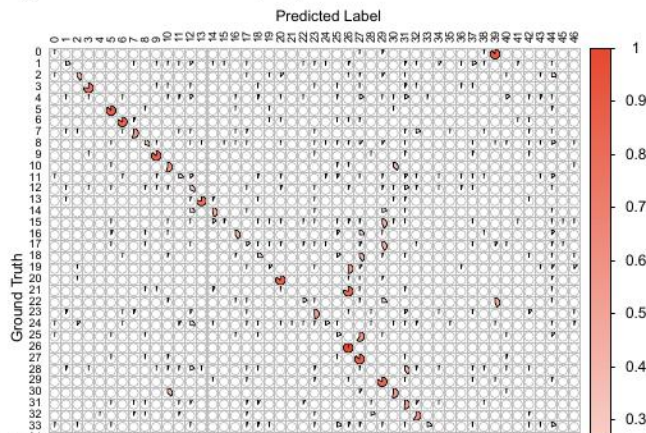
- Large VLMs show strong **zero-shot** ability, but adaptation to downstream tasks still needs many labels.
- Labeling is costly; recent work leverages **abundant unlabeled data** via pseudolabeling.
- Hard pseudolabels are often **inaccurate and imbalanced**, harming fine-tuning.



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Related Work

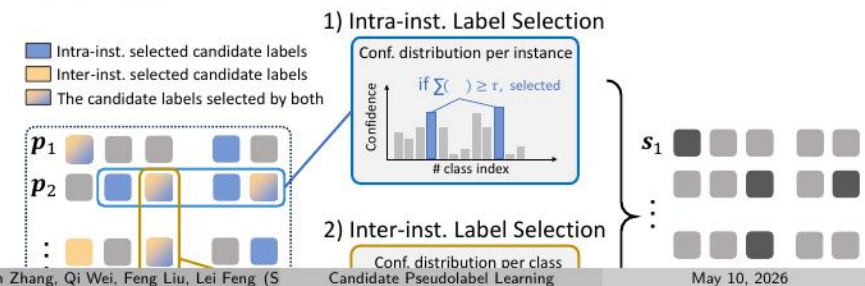
- Prompt tuning for VLMs: textual prompts and visual prompts.
- Pseudolabeling with VLMs: offline top-K, iterative update, consistency regularization.
- Challenge: Hard labels amplify **class bias** and **confirmation bias**.



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Method: Overview of Candidate Pseudolabel Learning (CPL)

- Goal: Fine-tune VLMs with **candidate pseudolabels** rather than a single hard label.
- Build a confidence-score matrix over unlabeled data; then select candidates by:
 - Intra-instance** selection (adaptive top-K by confidence threshold).
 - Inter-instance** selection (class-wise quantile filtering).
- Train with **partial-label learning** losses; iterate update of candidates and model.



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Method: Intra-instance Label Selection

- For unlabeled instance $\mathbf{x}_i$, obtain class scores $\mathbf{p}_i = (p_{i1}, \dots, p_{iC})^\top$ by softmax.
- Build S_i^{intra} by including labels (descending confidence) until cumulative score exceeds τ :

$$S_i^{\text{intra}} = \text{MinSize}\left(\left\{c \mid \sum_{c=1}^C p_{ic} \geq \tau\right\}\right).$$
- Adaptive threshold** τ from the distribution of $\hat{p}_i = \max_c p_{ic}$ over unlabeled data:

$$\tau = \text{Quantile}(\text{Sort}(\hat{\mathbf{p}}), \alpha), \quad \hat{\mathbf{p}} = (\hat{p}_1, \dots, \hat{p}_N).$$
- Larger α typically yields **larger candidate sets**, improving true-label inclusion for hard cases.

Method: Inter-instance Label Selection

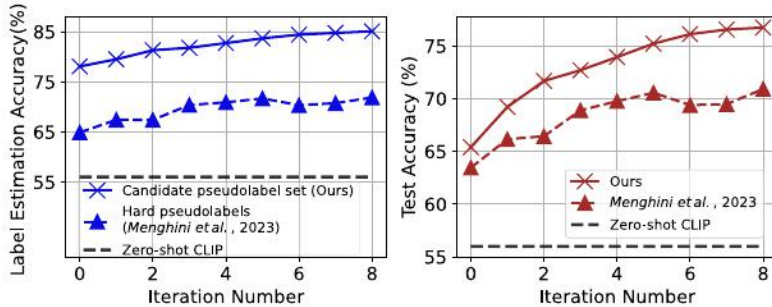
- Address class imbalance by **class-wise** filtering.
- For class c , collect $\mathbf{q}_c = (p_{1c}, \dots, p_{Nc})$; keep instance i for class c if it exceeds a **quantile**:

$$S_i^{\text{inter}} = \left\{c \mid p_{ic} > \text{Quantile}(\text{Sort}(\mathbf{q}_c), \beta)\right\}.$$
- Final candidate set per instance: $S_i = S_i^{\text{intra}} \cap S_i^{\text{inter}}$.
- Balances inclusion (intra) and **class balance** (inter) for robust supervision.

Learning with Candidate Pseudolabels and Iteration

- Convert candidate set S_i into target vector $\mathbf{s}_i \in \{0, 1\}^C$.
- Semi-supervised/TRZSL objective:

$$\mathcal{L} = \frac{1}{b_1} \sum_{i=1}^{b_1} L_{\text{ce}}(\mathbf{x}_i, \mathbf{y}_i) + \lambda \frac{1}{b_2} \sum_{i=1}^{b_2} L_{\text{p}}(\mathbf{x}_i, \mathbf{s}_i).$$
- Unsupervised objective: $\mathcal{L} = \frac{1}{b_2} \sum_i L_{\text{p}}(\mathbf{x}_i, \mathbf{s}_i)$.
- Iterate**: update model, regenerate candidates, and repeat (curriculum-style top- K_t per class).



Experimental Method and Setting

- Tasks: **UL**, **SSL** (2 labels/class), **TRZSL** (seen/unseen split; report harmonic mean).
- Backbones: CLIP ViT-B/32 (default), with textual/visual prompt tuning.
- Optimization: SGD, cosine schedule, 50 epochs; candidate update over T iterations.
- Candidate hyperparameters: α (for τ) and β (class-wise quantile).

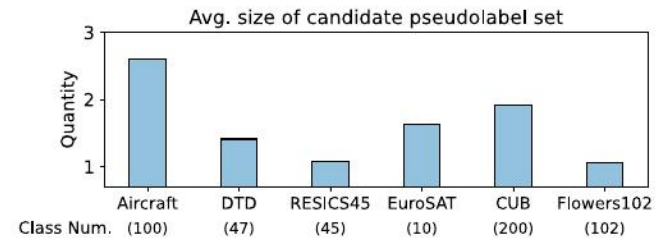


Figure 5: Average candidate-set size is often close to 1, limiting ambiguity while

Experimental Results (Parameter-Efficient UL)

- CPL modules can be plugged into existing pipelines (e.g., LaFTer) and **consistently improve** performance.

Table 1: Top-1 accuracy (%) with parameter-efficient tuning on six benchmarks (UL).

Method	Flowers102	UCF101	CIFAR100	EuroSAT	DTD	CALTECH101
CLIP	66.6	61.0	64.2	45.1	42.9	90.5
CLIP-PR	57.7	57.9	63.2	44.2	40.1	84.8
UPL	71.5	63.9	65.8	62.2	48.0	90.6
LaFTer	71.0	68.2	74.6	73.9	46.1	93.3
LaFTer + CPL	76.7	71.0	77.3	82.2	56.3	93.4

- CPL yields strong gains, especially on **EuroSAT** and **DTD**, where zero-shot CLIP is weaker.

Ablation: Losses and Strategy Effects

Table 2: DTD (Textual prompts, full unlabeled): effect of partial-label losses.

Method	SSL	UL	TRZSL
GRIP (hard)	56.07	46.09	65.30
CPL-CC	61.21	51.91	68.00
CPL-RC	60.21	51.58	67.95
CPL-LW	60.20	52.23	68.29

- CPL is **robust** across multiple partial-label losses and surpasses hard pseudolabeling.

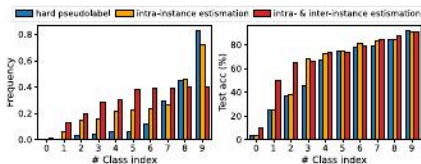
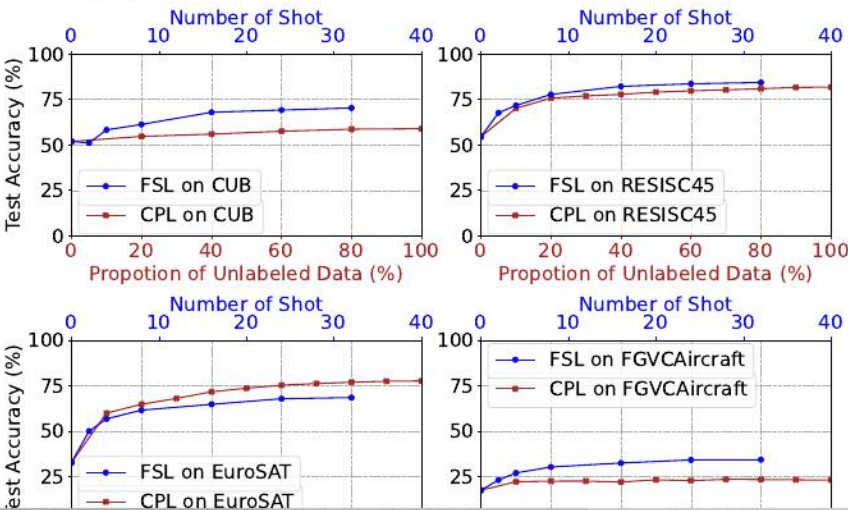


Figure 6: Strategy effects: balancing pseudolabel frequencies and boosting class-level accuracy.

How Much Unlabeled Data Helps?

- With more unlabeled data, gains **saturate**; few strong labels can still be highly valuable.



Deficiencies and Future Research

Current Limitations

- Candidate sets may still **miss the true label** when zero-shot scores are very unreliable.
- Sensitivity to α and β outside recommended ranges can reduce utilization or add ambiguity.
- Iterative regeneration adds **compute overhead**.

Future Directions

- Better **candidate-set priors**: structure- or retrieval-aware selection.
- **Bias-resistant** soft targets that avoid confirmation bias while using score magnitudes.
- Extending CPL to **dense** or **multimodal** tasks (detection, segmentation, retrieval).

Thank you!

- Key idea: **Candidate Pseudolabel Learning** balances inclusion and class equity.
- Consistent gains across tasks, prompts, and backbones.